KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 7 MATHEMATICS · 2026 YEARLY EXAM

Exam preparation booklet

Key ideas · worked examples · exam-style practice · answers

Multiple Choice
15 marks · 15 min

Calculations
32 marks · 32 min

Data and Probability
23 marks · 23 min

Name: _____

Teacher: _____

How to use this booklet

This booklet follows the **Year 7 Mathematics 2026 Yearly Exam** part by part. Every skill on the exam skills checklist has **key ideas**, a **worked example** and **practice questions** in the same style as the exam. Answers are at the back – try each question first, then check.

Part	Marks	Suggested time
A · Multiple Choice	15	15 minutes
B · Calculations	32	32 minutes
C · Data and Probability	23	23 minutes
Checking		5 minutes
Total	**70**	**75 minutes** + 5 min reading

Exam technique

- **No calculator.** Practise the written and mental methods in this booklet – you will need them.
- **Use your 5 minutes of reading time** to read every question and plan where to start.
- **About one minute per mark.** If you are stuck, move on and come back.
- **Multiple choice:** read all four options, cross out the ones you know are wrong, then circle **one** letter.
- **Show your working.** Marks are given for method, not just the final answer.
- **Bring:** black pen, pencil, ruler and eraser.
- **Check:** substitute answers back into equations; probabilities must be between 0 and 1.

Key words in exam questions

These instruction words tell you exactly what the marker is looking for.

Word	What you need to do	Example in the exam
Circle ONE	Choose the single best answer – circling two answers scores zero	Part A
Calculate … and provide working out	Write each step and the reason (e.g. angle sum of a triangle)	Part B Q1, Q2
State true or false	Write the word True or False for each part	Part B Q3
Convert	Change to the form asked for (mixed number or improper fraction)	Part B Q6
Write an algebraic expression	Use the letter given – no equals sign	Part B Q7
Solve by substitution	Replace each letter with its value and work out the answer	Part B Q8
Simplify	Collect like terms or multiply terms to write it more simply	Part B Q9
Solve	Find the value of the pronumeral	Part B Q11
Construct	Draw the graph or plot neatly with a ruler, title and labels	Part C Q12
List the sample space	Write every different outcome once, in curly brackets	Part C Q15

Self-check: how confident am I?

Tick one box for each skill before you start. Spend most of your revision time on the skills you tick "Help!". The **Booklet** column tells you where to find each skill.

Skill	Booklet	Exam Q	Got it	Nearly	Help!
Part A – Multiple Choice					
Know the meaning of mean, median, mode and range	A1	Q1, Q2, Q11	☐	☐	☐
Know the order of operations (brackets first, then × and ÷, then + and −)	A2	Q3	☐	☐	☐
Understand ascending and descending order	A3	Q4	☐	☐	☐
Name the place value of decimal digits (tenths, hundredths, thousandths)	A4	Q4 (2nd)	☐	☐	☐
Understand index notation, e.g. $11^3 = 11 \times 11 \times 11$	A5	Q5	☐	☐	☐
Know what a variable (pronumeral) is	A6	Q6	☐	☐	☐
Know that the angles in a triangle add to 180°	A7	Q7	☐	☐	☐
Identify the features of an isosceles triangle	A8	Q8	☐	☐	☐
Know what perimeter measures	A9	Q9	☐	☐	☐
Understand negative numbers on a number line	A10	Q10	☐	☐	☐
Know the sample space for rolling a die and what P(E) means	A11	Q12, Q13	☐	☐	☐
Classify angles (acute, right, obtuse, straight, reflex)	A12	Q14	☐	☐	☐
Know the angle relationships on parallel lines	A13	Q15	☐	☐	☐
Part B – Calculations					
Find a missing angle in a triangle using the angle sum	B1	Q1	☐	☐	☐
Find the exterior angle of a triangle	B2	Q2	☐	☐	☐
Identify equivalent fractions	B3	Q3	☐	☐	☐
Multiply decimals and fractions without a calculator	B4	Q4	☐	☐	☐
Add and subtract integers (negative numbers)	B5	Q5a, b, Q10ii	☐	☐	☐
Use the order of operations with brackets and integers	B6	Q5c	☐	☐	☐
Convert between improper fractions and mixed numbers	B7	Q6	☐	☐	☐
Write an algebraic expression from a worded statement	B8	Q7	☐	☐	☐
Substitute values into algebraic expressions	B9	Q8	☐	☐	☐
Simplify expressions by collecting like terms	B10	Q9i–iii	☐	☐	☐
Multiply algebraic terms, e.g. $3 \times 3y$	B11	Q9iv	☐	☐	☐
Use mental strategies for addition and multiplication	B12	Q10	☐	☐	☐
Solve one-step equations (+, −, ×, ÷)	B13	Q11i–iv	☐	☐	☐
Solve two-step equations, including ones with a fraction	B14	Q11v, vi	☐	☐	☐
Part C – Data and Probability					
Construct a dot plot from a list of data	C1	Q12i	☐	☐	☐
Read a dot plot to find the total and count values "less than" a number	C2	Q12ii, v	☐	☐	☐
Find the mode and range of a data set	C3	Q12iii, iv, Q13iii, iv	☐	☐	☐
Identify an outlier in a data set	C4	Q12vi	☐	☐	☐
Calculate the mean of a data set	C5	Q13i	☐	☐	☐
Find the median of a data set	C6	Q13ii	☐	☐	☐
Work out the new mean and median when a value is added	C7	Q13v, vi	☐	☐	☐
Read and interpret a column graph, including finding totals	C8	Q14i–iv	☐	☐	☐
Write a value from a graph as a fraction of a whole	C9	Q14v	☐	☐	☐
List the sample space for a spinner	C10	Q15a	☐	☐	☐
Calculate the probability of a simple event	C11	Q15b	☐	☐	☐
Find the complement of an event	C12	Q15c	☐	☐	☐

PART A

Multiple Choice

15 minutes | 15 marks

Key words, facts and definitions you need to recognise quickly. Questions 1–15 in the exam – circle **one** answer for each.

Skills in this section

- Mean, median, mode and range; order of operations
- Ascending and descending order; decimal place value
- Index notation and variables
- Triangles: angle sum and isosceles triangles
- Perimeter
- Negative numbers on a number line
- Sample space and P(E)
- Types of angles; angles on parallel lines

Number and data words

A1 Mean, median, mode and range

KEY IDEAS

Four words, four different jobs

Word	Meaning	Memory tip
Mean	Add all the values, then divide by how many values there are	the "average"
Median	The **middle** value when the data is in order	the median strip is in the middle of the road
Mode	The value that occurs **most** often	MOde = MOst
Range	Highest value − lowest value	how spread out the data is

WORKED EXAMPLE

For 3, 7, 4, 7, 9 → in order: 3, 4, **7**, 7, 9

Mean = $\frac{3+7+4+7+9}{5} = \frac{30}{5} = \mathbf{6}$ Median = **7** Mode = **7** Range = 9 − 3 = **6**

IN THE EXAM Part A, Q1, Q2 and Q11

Practice

(a) Which word means "the difference between the highest and lowest values"?

(b) Which word means "the middle value when the data is in order"?

(c) Which word means "add the values and divide by how many there are"?

(d) Which word means "the value that occurs most often"?

(e) For the data 2, 5, 5, 8, 10 find the mean, median, mode and range.

A2 Order of operations

KEY IDEAS

BIDMAS – brackets first

Order	Operation
1st	**B**rackets
2nd	**I**ndices (powers, e.g. 3^2)
3rd	**D**ivision and **M**ultiplication – working left to right
4th	**A**ddition and **S**ubtraction – working left to right

Multiplication does **not** always come before division – they have equal priority, so work from left to right. The same goes for + and −.

WORKED EXAMPLE

(a) $3 + 4 \times 5 = 3 + 20 =$ **23** (not $7 \times 5 = 35$)

(b) $(8 - 2) \times 3 = 6 \times 3 =$ **18** (c) $20 \div 4 \times 2 = 5 \times 2 =$ **10** (left to right)

IN THE EXAM Part A, Q3

Practice – calculate

(a) $6 + 2 \times 3$ **(b)** $10 - 4 \div 2$ **(c)** $(5 + 3) \times 2$ **(d)** $18 \div (2 + 4)$

(e) $4 \times 3 - 2 \times 5$ **(f)** $24 \div 6 \times 2$ **(g)** $5 + 3^2$ **(h)** $2 \times (9 - 4) + 1$

A3 Ascending and descending order

KEY IDEAS

Ascending goes up, descending goes down

Ascending = smallest to largest (like **a**scending a staircase). **Descending** = largest to smallest (like going **d**own).

Decimals: line up the decimal points and compare place by place (add zeros if it helps: 0.5 = 0.50).
Negatives: the further left on the number line, the smaller the number.

WORKED EXAMPLE

Order 0.5, 0.45, 0.405, 0.54 → write as 0.500, 0.450, 0.405, 0.540

Ascending: **0.405, 0.45, 0.5, 0.54** Descending: **0.54, 0.5, 0.45, 0.405**

IN THE EXAM Part A, Q4

Practice

(a) Write in ascending order: 12, 7, 31, 19, 3

(b) Write in descending order: 45, 54, 4, 405, 50

(c) Write in ascending order: 0.7, 0.07, 0.77, 0.707

(d) Write in descending order: −3, 5, 0, −8, 2

(e) Write in ascending order: $\frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}$

(f) What does *descending* order mean?

A4 Decimal place value

KEY IDEAS

Count places after the decimal point

1st decimal place = **tenths** · 2nd = **hundredths** · 3rd = **thousandths** · 4th = **ten-thousandths**

Hundreds	Tens	Units	.	Tenths	Hundredths	Thousandths	Ten-thousandths
		3	.	6	1	8	2
100	10	1		$\frac{1}{10}$	$\frac{1}{100}$	$\frac{1}{1000}$	$\frac{1}{10000}$

The names end in "**ths**" after the decimal point. Each place is 10 times smaller than the place to its left.

WORKED EXAMPLE

In 3.6182: the 6 is in the tenths place, the 1 is in the hundredths place, the 8 is in the thousandths place and the 2 is in the ten-thousandths place.

The value of the 8 is $\frac{8}{1000}$ = **0.008**

IN THE EXAM Part A, Q4 (second Q4)

Practice

(a) In 4.2759, what is the place value of the 7?

(b) In 4.2759, what is the place value of the 5?

(c) In 0.0436, which digit is in the hundredths place?

(d) What is the value of the 3 in 12.083?

(e) Write the number with 6 units, 4 tenths, 0 hundredths and 9 thousandths.

(f) What is the place value of the **third** decimal place?

A5 Index notation

KEY IDEAS

The index tells you how many times to multiply the base

In 5^3, 5 is the **base** and 3 is the **index** (power). $5^3 = 5 \times 5 \times 5 = 125$. It does **not** mean 5×3.

Power of 2 = **squared** · power of 3 = **cubed** · $x^4 = x \times x \times x \times x$

WORKED EXAMPLE

(a) $2^5 = 2 \times 2 \times 2 \times 2 \times 2 =$ **32** (b) $7 \times 7 \times 7 \times 7$ in index form = $\mathbf{7^4}$

IN THE EXAM Part A, Q5

Practice

(a) Write 6^3 in expanded form

(b) Write 9^5 in expanded form

(c) Write $4 \times 4 \times 4 \times 4 \times 4$ in index form

(d) Write $10 \times 10 \times 10$ in index form

(e) Evaluate 3^2

(f) Evaluate 2^4

(g) Evaluate 10^3

(h) Evaluate 5^3

A6 Variables

KEY IDEAS

A variable is a letter in place of a number

A **variable** (or pronumeral) is a letter that stands for a number. Its value can change (vary). $3x$ means $3 \times x$.

Word	Example	Meaning
Term	$4a$	a number, a letter, or numbers and letters multiplied together
Coefficient	the 4 in $4a$	the number in front of the variable
Constant	the 7 in $4a + 7$	a number on its own
Expression	$4a + 7$	terms joined by + or −, with **no** equals sign
Equation	$4a + 7 = 15$	has an **equals sign**

WORKED EXAMPLE

In $4a + 7$: the variable is a, the coefficient is **4**, the constant is **7**, and it is an **expression** (no equals sign).

IN THE EXAM Part A, Q6

Practice

(a) In $5m - 2$, what is the variable?

(b) In $5m - 2$, what is the coefficient?

(c) In $5m - 2$, what is the constant term?

(d) Is $2y + 1 = 9$ an expression or an equation?

(e) Which one is a variable: 7, x, +, = ?

(f) What does $6k$ mean?

Shapes and angles

A7 The angle sum of a triangle

KEY IDEAS

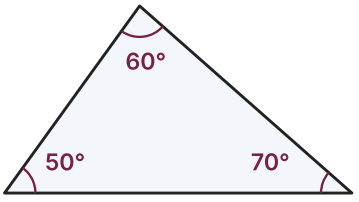
The three angles in a triangle add to 180°

Angle facts to know: **triangle** = 180° · **straight line** = 180° · **right angle** = 90° · **revolution** (full turn) = 360°.

WORKED EXAMPLE

Two angles of a triangle are 50° and 70°. Find the third angle.

180° − 50° − 70° = **60°** Check: 50 + 70 + 60 = 180 ✓



IN THE EXAM Part A, Q7

Practice

(a) Can 40°, 60° and 80° be the angles of a triangle? Explain.

(b) Can 90°, 45° and 55° be the angles of a triangle? Explain.

(c) Two angles of a triangle are 35° and 65°. Find the third angle.

(d) A right-angled triangle has an angle of 30°. Find the other angle.

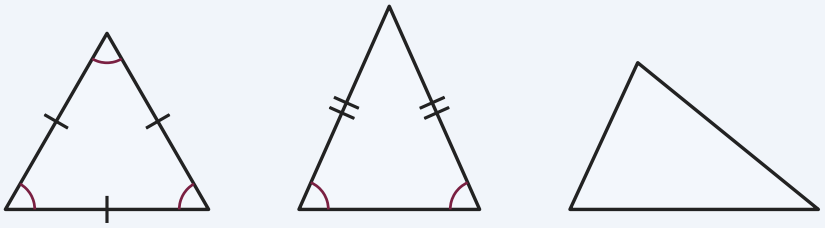
(e) Each angle in an equilateral triangle is equal. What size is each one?

A8 Isosceles triangles

KEY IDEAS

Isosceles = 2 equal sides and 2 equal angles

Matching marks (ticks) show sides of equal length; matching arcs show equal angles.



Equilateral Isosceles Scalene

Triangle	Sides	Angles
Equilateral	3 equal sides	3 equal angles (60° each)
Isosceles	**2 equal sides**	**2 equal angles** (opposite the equal sides)
Scalene	no equal sides	no equal angles

WORKED EXAMPLE

An isosceles triangle has a top (apex) angle of 40°. Find the two equal base angles.

180° − 40° = 140° → 140° ÷ 2 = **70° each**

IN THE EXAM Part A, Q8

Practice

(a) Name the triangle with sides 5 cm, 5 cm, 8 cm.

(b) Name the triangle with sides 6 cm, 6 cm, 6 cm.

(c) Name the triangle with sides 4 cm, 7 cm, 9 cm.

(d) An isosceles triangle has two base angles of 65°. Find the third angle.

(e) An isosceles triangle has an apex angle of 100°. Find each base angle.

(f) Which is a feature of an isosceles triangle: all sides equal, 2 equal sides, 3 equal angles, or no equal sides?

A9 Perimeter

KEY IDEAS

Perimeter = the distance around the outside

Add up the lengths of **all** the sides. Perimeter is a length, so the units are cm, m or km.

Measurement	What it measures	Units
Perimeter	the length around the outside of a shape	cm, m
Area	the amount of space covered by a flat shape	cm^2, m^2
Volume	the space taken up by a 3D object	cm^3, m^3
Capacity	the amount of liquid a container holds	mL, L

WORKED EXAMPLE

A rectangle is 8 cm long and 5 cm wide.

P = 8 + 5 + 8 + 5 = **26 cm** (or 2 × (8 + 5) = 26 cm)

IN THE EXAM Part A, Q9

Practice – find the perimeter

(a) A square with sides of 7 cm

(b) A rectangle 12 m long and 3 m wide

(c) A triangle with sides 5 cm, 6 cm and 9 cm

(d) A regular hexagon (6 equal sides) with sides of 4 cm

(e) Which measures the space covered by a shape: perimeter, area or volume?

(f) A rectangle has a perimeter of 20 cm and a length of 6 cm. Find its width.

A10 Negative numbers on a number line

KEY IDEAS

Left of zero is negative



Numbers to the **left** of zero are **negative**; numbers to the **right** are **positive**. The further left, the smaller the number, so −7 < −2. Adding moves right; subtracting moves left.

WORKED EXAMPLE

(a) Which is smaller, −5 or −2? −5 is further left, so **−5** is smaller.

(b) The temperature is −3 °C and rises 5 degrees: −3 + 5 = **2 °C**

IN THE EXAM Part A, Q10

Practice

(a) Which is larger: −4 or −9?

(b) Write < or >: −6 ___ 1

(c) Write < or >: −2 ___ −8

(d) What number is 3 to the left of 1?

(e) Write in ascending order: 3, −1, −5, 0, 2

(f) The temperature is −4 °C and falls 3 degrees. What is the new temperature?

A11 Sample space and P(E)

KEY IDEAS

The sample space lists every possible outcome

The **sample space** is the list of all possible outcomes, often written in curly brackets. Rolling a die: {1, 2, 3, 4, 5, 6}.

P(E) means "the probability that event E happens". $P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$

Probabilities go from **0** (impossible) to **1** (certain).

WORKED EXAMPLE

A die is rolled. Sample space = **{1, 2, 3, 4, 5, 6}**

P(even) = $\frac{3}{6} = \frac{1}{2}$ P(7) = **0** (impossible)

IN THE EXAM Part A, Q12 and Q13

Practice

(a) Write the sample space for tossing a coin.

(b) Write the sample space for choosing a letter from the word MATHS.

(c) A die is rolled. Find P(5).

(d) A die is rolled. Find P(a number less than 3).

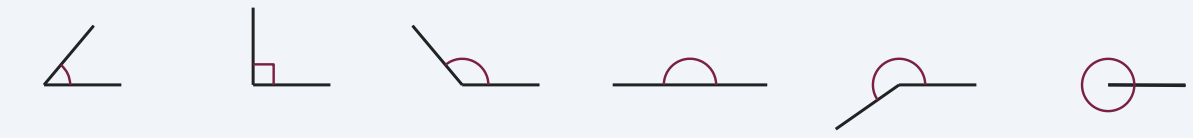
(e) A die is rolled. Find P(a number greater than 6).

(f) True or false: a probability can be 1.5.

A12 Types of angles

KEY IDEAS

Name an angle by its size



Acute	Right	Obtuse	Straight	Reflex	Revolution
less than 90°	exactly 90°	between 90° and 180°	exactly 180°	between 180° and 360°	exactly 360°

WORKED EXAMPLE

45° is **acute** · 135° is **obtuse** · 210° is **reflex** · 90° is a **right** angle

IN THE EXAM Part A, Q14

Practice – name each type of angle

(a) 72° **(b)** 90° **(c)** 165° **(d)** 180°

(e) 300° **(f)** 8° **(g)** 91° **(h)** 360°

A13 Angles on parallel lines

KEY IDEAS

Corresponding and alternate angles are equal; co-interior angles add to 180°

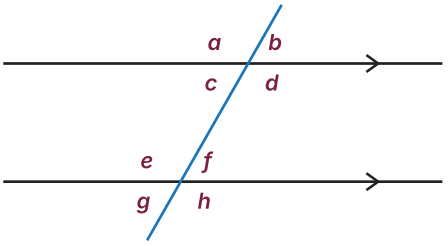
Pair	Shape to look for	Rule	Example in the diagram
Corresponding	F shape	equal	$a = e$
Alternate	Z shape	equal	$d = e$
Co-interior	C (or U) shape	add to 180°	$c + e = 180°$
Vertically opposite	X shape	equal	$a = d$

The arrows on the lines show they are parallel. *Corresponding* angles are in the **same position** at each intersection.

WORKED EXAMPLE

If $b = 65°$: its corresponding angle f **= 65°**

If $d = 115°$: its alternate angle e **= 115°**, and its co-interior angle $f = 180° - 115° =$ **65°**



IN THE EXAM Part A, Q15

Practice – use the diagram in the worked example. Treat each part separately

(a) Which angle corresponds to c?

(b) Which angle is alternate to c?

(c) Which angle is co-interior with d?

(d) If $a = 115°$, find e.

(e) If $c = 65°$, find f.

(f) If $d = 120°$, find f.

(g) Are corresponding angles on parallel lines supplementary, complementary or equal?

PART B

Calculations

32 minutes | 32 marks

Angles, fractions, decimals, integers, algebra and equations – all **without a calculator**. Questions 1–11 in Part B.

Skills in this section

- Missing and exterior angles in triangles
- Equivalent fractions; multiplying decimals and fractions
- Integers and the order of operations
- Improper fractions and mixed numbers
- Writing expressions; substitution
- Collecting like terms; multiplying terms
- Mental strategies
- One- and two-step equations

Angles

B1 Finding a missing angle in a triangle

KEY IDEAS

Angle sum of a triangle = 180°

Add the angles you know, then subtract from 180°. Write the reason: *angle sum of a triangle*. In a right-angled triangle, the square mark means 90°.

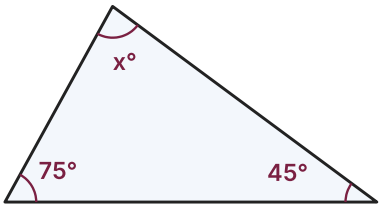
WORKED EXAMPLE

x + 75 + 45 = 180 (angle sum of a triangle)

x = 180 − 120

x = 60

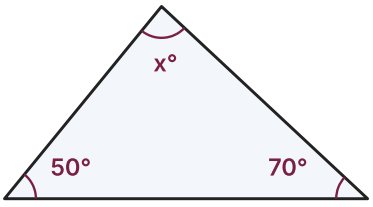
Check: 60 + 75 + 45 = 180 ✓



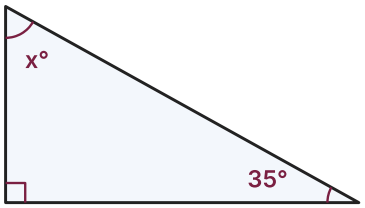
IN THE EXAM Part B, Q1 (2 marks)

Practice – find the value of *x*, giving a reason (diagrams not to scale)

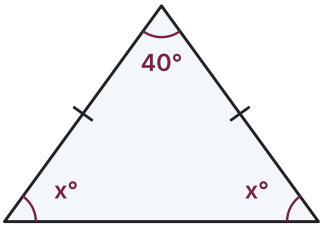
(a)



(b)



(c)



(d) Two angles of a triangle are 110° and 25°. Find the third angle.

B2 The exterior angle of a triangle

KEY IDEAS

Exterior angle = sum of the two interior opposite angles

An **exterior angle** is made when one side of the triangle is extended. It equals the sum of the two interior angles at the **other** two corners.

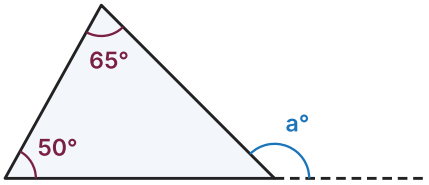
You can also use angles on a straight line: exterior angle + the interior angle next to it = 180°.

WORKED EXAMPLE

a = 50 + 65 (exterior angle of a triangle)

a **= 115**

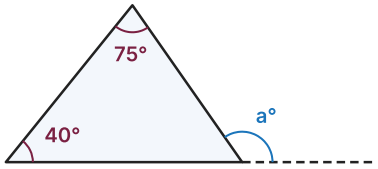
Check: third interior angle = 180 − 50 − 65 = 65°, and 180 − 65 = 115° ✓



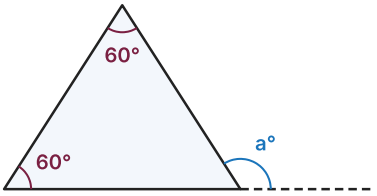
IN THE EXAM Part B, Q2 (2 marks)

Practice – find the size of the exterior angle *a* (diagrams not to scale)

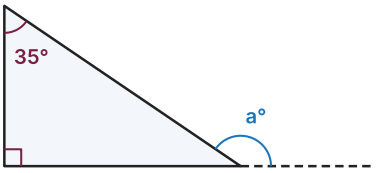
(a)



(b)



(c)



(d) An exterior angle of a triangle is 140°. One of the interior opposite angles is 85°. Find the other interior opposite angle.

Fractions, decimals and integers

B3 Equivalent fractions

KEY IDEAS

Multiply or divide the top and bottom by the same number

$\frac{3}{5} = \frac{3 \times 2}{5 \times 2} = \frac{6}{10}$ – equivalent fractions are equal in value.

To test two fractions: find what the numerator was multiplied by, and check the denominator was multiplied by the **same** number. Or cross-multiply: $\frac{a}{b} = \frac{c}{d}$ when $a \times d = b \times c$.

WORKED EXAMPLE

(a) $\frac{2}{5} = \frac{8}{20}$? 2 × 4 = 8 and 5 × 4 = 20 → **True**

(b) $\frac{4}{7} = \frac{12}{28}$? 4 × 3 = 12 but 7 × 3 = 21, not 28 → **False**

IN THE EXAM Part B, Q3 (2 marks)

Practice – true or false?

(a) $\frac{1}{3} = \frac{4}{12}$ **(b)** $\frac{2}{3} = \frac{6}{12}$ **(c)** $\frac{5}{8} = \frac{15}{24}$

(d) $\frac{3}{5} = \frac{9}{20}$ **(e)** $\frac{7}{10} = \frac{21}{30}$ **(f)** $\frac{4}{9} = \frac{16}{27}$

(g) Fill the gap: $\frac{3}{4} = \frac{?}{20}$ **(h)** Fill the gap: $\frac{2}{7} = \frac{10}{?}$

B4 Multiplying decimals and fractions

KEY IDEAS

No calculator – use these methods

Decimals: multiply without the decimal point, then put back the same number of decimal places as the question. 0.3 × 4 → 3 × 4 = 12 → **1.2**

Fractions: multiply the numerators, multiply the denominators, then simplify. $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$. "**of**" means ×.

WORKED EXAMPLE

(a) 0.4 × 3: 4 × 3 = 12, one decimal place → **1.2** (b) 0.02 × 6: 2 × 6 = 12, two decimal places → **0.12**

(c) $\frac{2}{3} \times \frac{1}{5} = \frac{2 \times 1}{3 \times 5} = \mathbf{\frac{2}{15}}$ (d) $\frac{1}{4} \times \frac{2}{3} = \frac{2}{12} = \mathbf{\frac{1}{6}}$

IN THE EXAM Part B, Q4 (2 marks)

Practice – calculate

(a) 0.3 × 2 **(b)** 0.4 × 5 **(c)** 0.7 × 3 **(d)** 0.05 × 4

(e) $\frac{1}{2} \times \frac{1}{4}$ **(f)** $\frac{2}{5} \times \frac{1}{3}$ **(g)** $\frac{3}{4} \times \frac{2}{3}$ **(h)** $\frac{1}{3}$ of 18

B5 Adding and subtracting integers

KEY IDEAS

Use a number line



Adding a positive → move **right**. Subtracting a positive → move **left**.

Two signs together: **+ −** or **− +** becomes **−**; **− −** becomes **+**. e.g. 4 − (−2) = 4 + 2 = 6

WORKED EXAMPLE

(a) −8 + 11: start at −8, move 11 right → **3** (b) −3 − 6: start at −3, move 6 left → **−9**

(c) 4 − (−2) = 4 + 2 = **6** (d) −7 + 20 = **13**

IN THE EXAM Part B, Q5(a), Q5(b) and Q10(ii)

Practice – calculate

(a) −6 + 9 **(b)** −12 + 5 **(c)** −2 − 7 **(d)** 3 − 10

(e) −4 + 15 **(f)** 5 − (−3) **(g)** −9 − (−4) **(h)** −1 − 1

B6 Order of operations with brackets and integers

KEY IDEAS

Brackets first – even when there are negatives inside

Work out the brackets first, then multiply or divide, then add or subtract.

Multiplying and dividing integers: same signs → positive answer; different signs → negative answer. $3 \times (-7) = -21$, $(-4) \times (-5) = 20$

WORKED EXAMPLE

(a) $6 \times (-4 + 7) = 6 \times 3 = \mathbf{18}$ (b) $3 \times (2 - 9) = 3 \times (-7) = \mathbf{-21}$

(c) $-10 + 3 \times 4 = -10 + 12 = \mathbf{2}$ (× before +)

IN THE EXAM Part B, Q5(c)

Practice – calculate

(a) 5 × (−2 + 6) **(b)** 4 × (−1 + 3) **(c)** (−8 + 10) × 7 **(d)** 2 × (3 − 8)

(e) −11 + 2 × 4 **(f)** 9 − (−5 + 2) **(g)** (−6 − 4) ÷ 2 **(h)** 3 × (−5) + 20

B7 Improper fractions and mixed numbers

KEY IDEAS

Divide to make a mixed number; multiply and add to go back

Improper → mixed: divide the numerator by the denominator. The answer is the whole number and the remainder goes over the denominator. $\frac{11}{4}$: 11 ÷ 4 = 2 remainder 3 → $2\frac{3}{4}$

Mixed → improper: whole number × denominator + numerator, over the same denominator. $3\frac{2}{5} = \frac{3 \times 5 + 2}{5} = \frac{17}{5}$

WORKED EXAMPLE

(a) $\frac{7}{3}$: 7 ÷ 3 = 2 remainder 1 → $\mathbf{2\frac{1}{3}}$ (b) $2\frac{3}{4} = \frac{2 \times 4 + 3}{4} = \mathbf{\frac{11}{4}}$

IN THE EXAM Part B, Q6 (2 marks)

Practice – convert

(a) $\frac{5}{2}$ to a mixed number

(b) $\frac{9}{4}$ to a mixed number

(c) $\frac{16}{5}$ to a mixed number

(d) $\frac{23}{6}$ to a mixed number

(e) $1\frac{2}{3}$ to an improper fraction

(f) $3\frac{1}{2}$ to an improper fraction

(g) $2\frac{2}{5}$ to an improper fraction

(h) $5\frac{3}{4}$ to an improper fraction

Algebra

B8 Writing algebraic expressions

KEY IDEAS

Turn the words into symbols

Words	Algebra
the sum of the number and 4 / 4 more than the number	$x + 4$
the difference between the number and 2 / the number minus 2	$x - 2$
2 less than the number	$x - 2$ (**not** $2 - x$)
the product of 6 and the number / 6 times the number	$6x$
the number divided by 3	$\frac{x}{3}$
double the number / the number squared	$2x$ / x^2

Write numbers in front of letters ($6x$, not $x6$) and leave out the × sign.

WORKED EXAMPLE

(a) 7 more than the number → $\boldsymbol{x + 7}$ (b) the number divided by 4 → $\boldsymbol{\frac{x}{4}}$ (c) twice the number, minus 1 → $\boldsymbol{2x - 1}$

IN THE EXAM Part B, Q7 (2 marks)

Practice – write an expression, using x for the number

(a) 5 more than the number

(b) the number decreased by 8

(c) 4 times the number

(d) the number divided by 6

(e) the product of the number and 10

(f) 3 less than double the number

(g) the sum of the number and 9, all multiplied by 2

(h) the number squared

B9 Substitution

KEY IDEAS

Replace each letter with its value

ab means $a \times b$ · $3a$ means $3 \times a$ · then use the order of operations (× before +).

"Solve by substitution" in the exam means **work out the value** of the expression.

WORKED EXAMPLE

If $a = 5$, $b = 2$ and $c = 3$:

(a) $a + b + c = 5 + 2 + 3 =$ **10** (b) $bc + 4 = 2 \times 3 + 4 = 6 + 4 =$ **10** (c) $3a - b = 3 \times 5 - 2 =$ **13**

IN THE EXAM Part B, Q8 (2 marks)

Practice – find the value of each expression when $p = 6$, $q = 2$ and $r = 5$

(a) $p + q + r$ **(b)** $p - q$ **(c)** $4r$ **(d)** pq

(e) $qr + 1$ **(f)** $p + 2r$ **(g)** $pr - q$ **(h)** $p \div q$

B10 Collecting like terms

KEY IDEAS

Only like terms can be added or subtracted

Like terms have exactly the same letters: ab and $3ab$ are like terms; ab and a are not. Add or subtract the numbers in front and keep the letters. xy means $1xy$. Numbers on their own are like terms with each other.

WORKED EXAMPLE

(a) $5m + 3m =$ **$8m$** (b) $2xy + xy + 4xy =$ **$7xy$** (c) $8p + 5 - 3p = 8p - 3p + 5 =$ **$5p + 5$**

IN THE EXAM Part B, Q9(i)–(iii)

Practice – simplify

(a) $3a + 4a$ **(b)** $9k - 2k$ **(c)** $6t + t$ **(d)** $xy + 3xy + 5xy$

(e) $7m + 2 - 4m$ **(f)** $5g + 3h + 2g$ **(g)** $12n - 5 - 2n + 1$ **(h)** $4ab - ab$

B11 Multiplying algebraic terms

KEY IDEAS

Multiply the numbers, then write the letters together

$4 \times 2y = 8y$ · $3a \times 2b = 6ab$ · $a \times a = a^2$

Leave out the × sign and write the letters in alphabetical order. $3 \times 3y$ is **$9y$**, not $33y$.

WORKED EXAMPLE

(a) $5 \times 2k =$ **$10k$** (b) $2m \times 3n =$ **$6mn$** (c) $4x \times x =$ **$4x^2$**

IN THE EXAM Part B, Q9(iv)

Practice – simplify

(a) $2 \times 4x$ **(b)** $6 \times 5p$ **(c)** $3a \times 2$ **(d)** $4m \times 3n$

(e) $a \times b$ **(f)** $2y \times 7y$ **(g)** $5 \times 3gh$ **(h)** $10 \times 10d$

B12 Mental strategies for addition and multiplication

KEY IDEAS

Make the numbers friendly

Strategy	Example
Jump / split – add the tens, then the units	37 + 25 = 37 + 20 + 5 = 62
Compensation – round, then adjust	48 + 19 = 48 + 20 − 1 = 67
Make tens – pair numbers that add to 10, 20, …	13 + 8 + 7 + 2 = (13 + 7) + (8 + 2) = 30
Multiples of 10 and 100	6 × 300 = 6 × 3 × 100 = 1800
Split and multiply	23 × 8 = 20 × 8 + 3 × 8 = 160 + 24 = 184
× 9 = × 10, then subtract once	15 × 9 = 150 − 15 = 135

WORKED EXAMPLE

(a) −5 + 21 = 21 − 5 = **16** (b) 14 + 7 + 16 + 3 = (14 + 16) + (7 + 3) = 30 + 10 = **40**

(c) 8 × 500 = 8 × 5 × 100 = **4000** (d) 32 × 9 = 320 − 32 = **288**

IN THE EXAM Part B, Q10 (5 marks)

Practice – calculate mentally

(a) 7 + 35 **(b)** −4 + 17 **(c)** 15 + 9 + 5 + 11

(d) 48 + 29 **(e)** 8 × 300 **(f)** 6 × 500

(g) 24 × 9 **(h)** 13 × 6 **(i)** 31 × 4

Equations

B13 One-step equations

KEY IDEAS

Undo what has been done to the pronumeral

Operation	+	−	×	÷
Inverse (undo)	−	+	÷	×

Do the inverse operation to **both sides** so the pronumeral is on its own. Check by substituting your answer back in.

WORKED EXAMPLE

(a) $x + 7 = 15 \rightarrow x = 15 - 7 \rightarrow \boldsymbol{x = 8}$ (b) $h - 3 = 9 \rightarrow h = 9 + 3 \rightarrow \boldsymbol{h = 12}$

(c) $4n = 28 \rightarrow n = 28 \div 4 \rightarrow \boldsymbol{n = 7}$ (d) $\frac{m}{5} = 2 \rightarrow m = 2 \times 5 \rightarrow \boldsymbol{m = 10}$

IN THE EXAM Part B, Q11(i)–(iv)

Practice – solve

(a) $a + 6 = 10$ **(b)** $y + 11 = 20$ **(c)** $p - 5 = 8$ **(d)** $w - 7 = 7$

(e) $3x = 21$ **(f)** $6m = 48$ **(g)** $\frac{k}{2} = 7$ **(h)** $\frac{n}{5} = 4$

B14 Two-step equations, including fractions

KEY IDEAS

Undo in reverse order

Undo **+ or –** first, then **× or ÷**.

If the **whole top** of a fraction is divided, such as $\frac{r - 3}{2}$, multiply both sides by the denominator **first**, then undo the + or −.

WORKED EXAMPLE

(a) $3x + 5 = 26$ **−5** $3x = 21$ **÷3** $\boldsymbol{x = 7}$

(b) $\frac{p + 4}{3} = 5$ **×3** $p + 4 = 15$ **−4** $\boldsymbol{p = 11}$ Check: (11 + 4) ÷ 3 = 5 ✓

IN THE EXAM Part B, Q11(v) and Q11(vi)

Practice – solve, showing every step

(a) $2x + 3 = 11$ **(b)** $4m + 1 = 21$ **(c)** $5k - 2 = 13$ **(d)** $3y - 6 = 12$

(e) $\frac{a + 2}{3} = 4$ **(f)** $\frac{n - 5}{2} = 6$ **(g)** $\frac{w + 1}{5} = 3$ **(h)** $\frac{t - 4}{3} = 2$

PART C

Data and Probability

23 minutes | 23 marks

Dot plots, summary statistics, column graphs and simple probability. Questions 12–15 in Part C.

Skills in this section

- Constructing and reading dot plots
- Mode, range and outliers
- Mean and median, and adding a new value
- Column graphs; writing a value as a fraction of a whole
- Sample space, probability and complementary events

Dot plots

C1 Constructing a dot plot

KEY IDEAS

One dot for every data value

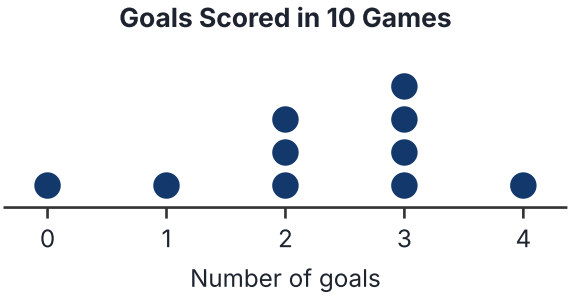
1. Find the smallest and largest values. Draw a number line covering them, **evenly spaced** – include values that have no dots.
2. Cross off each value as you go (or make a tally first) and draw **one dot** above its number. Stack dots evenly in neat columns.
3. Add a title and a label under the axis.
4. Check: the number of dots = the number of data values.

WORKED EXAMPLE

Goals scored: 3, 1, 2, 3, 4, 2, 3, 0, 3, 2

Goals	0	1	2	3	4
Frequency	1	1	3	4	1

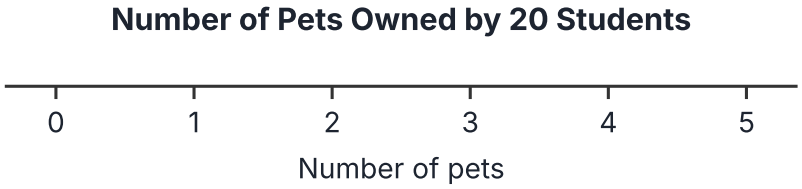
Check: 1 + 1 + 3 + 4 + 1 = 10 dots ✓



IN THE EXAM Part C, Q12(i)

Practice – the number of pets owned by 20 students: 1, 0, 2, 1, 3, 1, 2, 0, 1, 4, 2, 1, 0, 1, 2, 3, 1, 2, 5, 1

Construct a dot plot of this data on the axis below. (You will use it again in C2–C4.)



C2 Reading a dot plot: totals and "less than"

KEY IDEAS

Count the dots that match the words

Phrase	Means	Includes 4?
less than 4	0, 1, 2, 3	**No**
more than 4	5, 6, …	**No**
at least 4	4, 5, 6, …	**Yes**

The **total** (e.g. how many games were played) = the total number of dots.

WORKED EXAMPLE

Using the goals dot plot in C1:

Games played: **10** Fewer than 3 goals: 1 + 1 + 3 = **5 games** More than 2 goals: 4 + 1 = **5 games**

IN THE EXAM Part C, Q12(ii) and Q12(v)

Practice – use your pets dot plot from C1

(a) How many students were surveyed?

(b) How many students own fewer than 2 pets?

(c) How many students own more than 2 pets?

(d) How many students own at least 2 pets?

(e) How many students own exactly 1 pet?

Summary statistics

C3 Mode and range

KEY IDEAS

Most common, and the spread

Mode = the value that occurs most often – on a dot plot, the **tallest stack**. There can be two modes, or no mode.

Range = highest value − lowest value. Give one number, not "1 to 9".

WORKED EXAMPLE

Goals dot plot (C1): mode = **3 goals** (tallest stack); range = 4 − 0 = **4**

Data 6, 2, 9, 2, 5, 7: mode = **2**; range = 9 − 2 = **7**

IN THE EXAM Part C, Q12(iii), Q12(iv), Q13(iii) and Q13(iv)

Practice – find the mode and range

(a) Your pets dot plot from C1

(b) 8, 3, 5, 3, 9, 4, 3

(c) 12, 15, 11, 15, 20, 12

(d) 4, 7, 1, 9

C4 Outliers

KEY IDEAS

An outlier is far away from the rest of the data

An **outlier** is a value that is much larger or much smaller than the others. On a dot plot it sits on its own, with a big gap.

Not every data set has an outlier. Answer with "**Yes, the outlier is …**" or "**No, there is no outlier**".

WORKED EXAMPLE

4, 5, 5, 6, 4, 18, 5 → the other values are 4 to 6, so **yes, 18 is an outlier**.

IN THE EXAM Part C, Q12(vi)

Practice – is there an outlier? If so, what is it?

(a) 7, 8, 6, 7, 25, 8

(b) 3, 4, 5, 6, 7, 8

(c) 21, 19, 2, 20, 22

(d) 50, 52, 49, 51, 80, 48

C5 The mean

KEY IDEAS

Add them up, divide by how many

Mean = $\frac{\text{sum of the values}}{\text{number of values}}$. Without a calculator, add in friendly pairs (e.g. 3 + 7 = 10). The mean can be a decimal.

WORKED EXAMPLE

(a) 4, 6, 3, 5, 7: sum = 25, 5 values → $\frac{25}{5}$ = **5** (b) 2, 3, 4, 5: sum = 14, 4 values → $\frac{14}{4}$ = **3.5**

IN THE EXAM Part C, Q13(i)

Practice – find the mean

(a) 3, 5, 7

(b) 2, 4, 6, 8

(c) 1, 3, 4, 6, 6

(d) 10, 12, 14, 16, 18

(e) 5, 1, 2, 4

(f) 3, 4, 4, 5, 6, 8

(g) 1, 2, 2, 5

C6 The median

KEY IDEAS

Order first, then find the middle

1. Write the values in order from smallest to largest.
2. Cross off one value from each end until you reach the middle.
3. **Odd** number of values → one middle value. **Even** number → halfway between the two middle values.

WORKED EXAMPLE

(a) 7, 2, 9, 4, 6 → 2, 4, **6**, 7, 9 → median = **6**

(b) 5, 1, 8, 3 → 1, **3, 5**, 8 → median = $\frac{3 + 5}{2}$ = **4**

IN THE EXAM Part C, Q13(ii)

Practice – find the median

(a) 3, 9, 5
(b) 8, 2, 6, 4, 10
(c) 4, 1, 7, 3, 5, 2, 6
(d) 2, 8, 4, 6
(e) 10, 3, 7, 1, 9, 5
(f) 6, 2, 3, 6, 1, 4, 5, 2, 3

C7 Adding a new value: the new mean and median

KEY IDEAS

One more value → recount, re-order, recalculate

New mean = (old sum + new value) ÷ (old number of values + 1).
New median: put the new value in order with the rest and find the new middle. If there is now an even number of values, find halfway between the middle two.

A large new value pulls the mean up a lot, but the median only moves a little.

WORKED EXAMPLE

Data 2, 4, 4, 5, 5 (sum 20, mean 4, median 4). The value 10 is added.

New mean = $\frac{20 + 10}{6} = \frac{30}{6} = \mathbf{5}$ New median: 2, 4, **4, 5**, 5, 10 → $\frac{4 + 5}{2} = \mathbf{4.5}$

IN THE EXAM Part C, Q13(v) and Q13(vi)

Practice

1. Data: 3, 5, 6, 6, 8, 2, 5

(a) Find the mean.
(b) Find the median.
(c) The value 13 is added. Find the new mean.
(d) Find the new median.

2. Data: 4, 1, 3, 2, 5

(a) Find the mean.
(b) Find the median.
(c) The value 9 is added. Find the new mean.
(d) Find the new median.

Column graphs

C8 Reading a column graph

KEY IDEAS

Read across from the top of each column

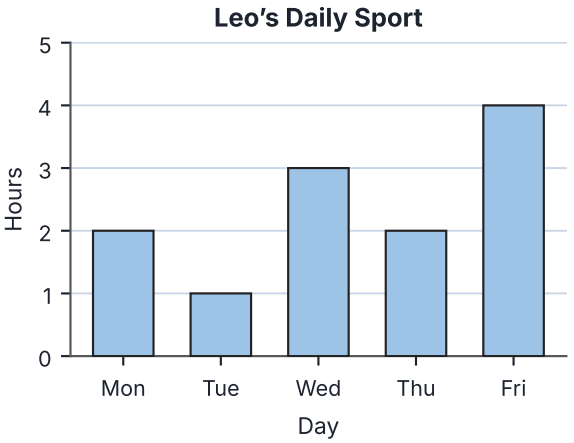
Check the scale on the vertical axis first. Most = tallest column; least = shortest column. **Total** = add the heights of all the columns. Include units (e.g. hours) in your answer.

WORKED EXAMPLE

Wednesday: **3 hours**

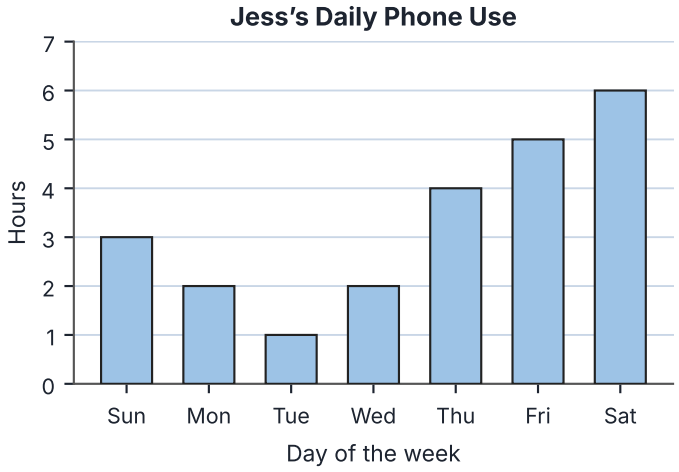
Most sport: **Friday** (4 hours)

Total for the week: 2 + 1 + 3 + 2 + 4 = **12 hours**



IN THE EXAM Part C, Q14(i)–(iv)

Practice



(a) How many hours did Jess use her phone on Tuesday?

(b) How many hours on Thursday?

(c) On which day did she use her phone the most?

(d) On which day did she use it the least?

(e) How many hours did she use her phone over the whole week?

(f) How many more hours on Saturday than Monday?

C9 Writing a value as a fraction of a whole

KEY IDEAS

Fraction = part ÷ whole, then simplify

Write $\frac{\text{part}}{\text{whole}}$, then divide the top and bottom by their highest common factor. Know the whole:

Whole	Size
1 day	24 hours
1 hour	60 minutes
1 week	7 days
1 year	12 months

WORKED EXAMPLE

(a) 6 hours of a day: $\frac{6}{24} = \frac{\mathbf{1}}{\mathbf{4}}$ (÷ 6) (b) 20 minutes of an hour: $\frac{20}{60} = \frac{\mathbf{1}}{\mathbf{3}}$ (÷ 20)

IN THE EXAM Part C, Q14(v)

Practice – write as a fraction in simplest form

(a) Jess's phone use on Saturday, as a fraction of the day

(b) Jess's phone use on Monday, as a fraction of the day

(c) Jess's phone use on Thursday, as a fraction of the day

(d) 8 hours of sleep, as a fraction of a day

(e) 45 minutes, as a fraction of an hour

(f) 3 days, as a fraction of a week

Probability

C10 Sample space for a spinner

KEY IDEAS

List each different outcome once

The **sample space** lists every **different** outcome once, in curly brackets – even if a colour appears on several sections.

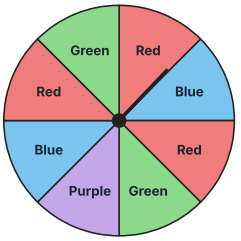
Count the sections later, when you work out probabilities.

WORKED EXAMPLE

Spinner A has 8 equal sections.

Sample space = **{red, blue, green, purple}**

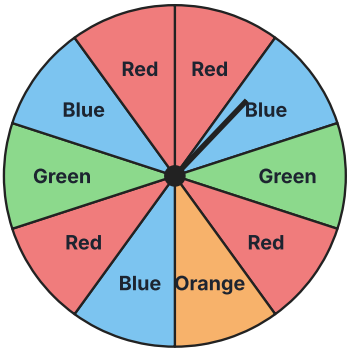
Not {red, red, red, blue, ...} – each outcome is listed once.



Spinner A

IN THE EXAM Part C, Q15(a)

Practice



Spinner B (10 equal sections)

(a) List the sample space for Spinner B.

(b) List the sample space for a spinner numbered 1 to 5.

(c) List the sample space for choosing a letter from the word BANANA.

(d) List the sample space for rolling a die and recording odd or even.

C11 Probability of a simple event

KEY IDEAS

P(E) = favourable outcomes ÷ total outcomes

$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$. For a spinner, count **sections**, not colours. Simplify your fraction.

Impossible = 0 · certain = 1 · all probabilities are between 0 and 1.

WORKED EXAMPLE

Spinner A (8 sections): P(red) = $\frac{3}{8}$ P(green) = $\frac{2}{8} = \frac{1}{4}$ P(yellow) = **0**

IN THE EXAM Part C, Q15(b)

Practice – use Spinner B from C10 for (a)–(f)

(a) P(red)

(b) P(blue)

(c) P(green)

(d) P(orange)

(e) P(red or blue)

(f) P(white)

(g) A die is rolled. Find P(3).

(h) A die is rolled. Find P(an odd number).

C12 Complementary events

KEY IDEAS

The complement of E is "not E"

The **complement** of an event is everything in the sample space that is **not** that event. The complement of spinning red on Spinner A is spinning **blue, green or purple**.

An event and its complement add to 1: **P(not E) = 1 – P(E)**

WORKED EXAMPLE

Spinner A: complement of spinning red = **spinning blue, green or purple**

P(not red) = $1 - \frac{3}{8} = \frac{5}{8}$ (check: 5 of the 8 sections are not red ✓)

IN THE EXAM Part C, Q15(c)

Practice – describe the complement, then find its probability

(a) Spinning blue on Spinner B

(b) Spinning orange on Spinner B

(c) Rolling a 6 on a die

(d) Rolling an even number on a die

(e) Tossing a head on a coin

(f) The probability of rain tomorrow is 0.3. What is the probability of no rain?

The night before and on the day

Formulas and facts to know by heart

- Mean = sum ÷ number of values; median = middle; mode = most; range = highest − lowest
- BIDMAS: brackets, indices, ÷ and × (left to right), + and − (left to right)
- $5^3 = 5 \times 5 \times 5$
- Triangle angle sum = 180°; straight line = 180°; revolution = 360°
- Exterior angle = sum of the two interior opposite angles
- Parallel lines: corresponding and alternate angles are equal; co-interior angles add to 180°
- Isosceles: 2 equal sides and 2 equal angles
- − − becomes +; + − becomes −
- Mixed → improper: whole × denominator + numerator
- P(E) = favourable ÷ total; P(not E) = 1 − P(E)

Common mistakes to avoid

- Reading 11^3 as 11×3
- Doing + before × ($3 + 4 \times 5$ is 23, not 35)
- Mixing up ascending and descending
- $-5 - 4 = -1$ (it is -9)
- Writing “the difference between the number and 5” as $5 - x$
- $3 \times 3y = 33y$ (it is $9y$)
- Subtracting 3 before multiplying by 2 in $\frac{r - 3}{2} = 8$
- Finding the median without ordering the data first
- Counting 4 in “less than 4”
- Listing a colour more than once in a sample space
- Counting colours instead of sections for a probability

Your 75-minute plan

0–15 min	15–47 min	47–70 min	70–75
Part A Multiple Choice	Part B Calculations	Part C Data and Probability	Check

Use the 5 minutes of reading time before the exam starts to read every question. Keep the last 5 minutes to go back to skipped questions and check your answers.

Answers

Part A · Multiple Choice

A1 (a) range (b) median (c) mean (d) mode (e) mean = 30 ÷ 5 = 6, median = 5, mode = 5, range = 10 − 2 = 8

A2 (a) 12 (b) 8 (c) 16 (d) 3 (e) 12 − 10 = 2 (f) 4 × 2 = 8 (g) 5 + 9 = 14 (h) 2 × 5 + 1 = 11

A3 (a) 3, 7, 12, 19, 31 (b) 405, 54, 50, 45, 4 (c) 0.07, 0.7, 0.707, 0.77 (d) 5, 2, 0, −3, −8 (e) $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ (f) largest to smallest

A4 (a) hundredths (b) thousandths (c) 4 (d) 3 thousandths = 0.003 (e) 6.409 (f) thousandths

A5 (a) 6 × 6 × 6 (b) 9 × 9 × 9 × 9 × 9 (c) 4^5 (d) 10^3 (e) 9 (f) 16 (g) 1000 (h) 125

A6 (a) m (b) 5 (c) −2 (d) an equation (it has an equals sign) (e) x (f) 6 × k

A7 (a) yes – 40 + 60 + 80 = 180 (b) no – 90 + 45 + 55 = 190 (c) 80° (d) 60° (e) 60°

A8 (a) isosceles (b) equilateral (c) scalene (d) 180 − 130 = 50° (e) (180 − 100) ÷ 2 = 40° (f) 2 equal sides

A9 (a) 28 cm (b) 30 m (c) 20 cm (d) 24 cm (e) area (f) 4 cm

A10 (a) −4 (b) < (c) > (d) −2 (e) −5, −1, 0, 2, 3 (f) −7 °C

A11 (a) {H, T} (b) {M, A, T, H, S} (c) $\frac{1}{6}$ (d) $\frac{2}{6} = \frac{1}{3}$ (e) 0 (f) false – probabilities are between 0 and 1

A12 (a) acute (b) right (c) obtuse (d) straight (e) reflex (f) acute (g) obtuse (h) revolution

A13 (a) g (b) f (c) f (d) 115° (corresponding) (e) 65° (alternate) (f) 60° (co-interior) (g) equal

Part B · Calculations

B1 (a) 60° (b) 55° (c) 70° (d) 45° – reason: angle sum of a triangle

B2 (a) 40 + 75 = 115° (b) 120° (c) 90 + 35 = 125° (d) 140 − 85 = 55°

B3 (a) true (b) false (c) true (d) false (e) true (f) false (g) 15 (h) 35

B4 (a) 0.6 (b) 2 (c) 2.1 (d) 0.2 (e) $\frac{1}{8}$ (f) $\frac{2}{15}$ (g) $\frac{6}{12} = \frac{1}{2}$ (h) 6

B5 (a) 3 (b) −7 (c) −9 (d) −7 (e) 11 (f) 8 (g) −5 (h) −2

B6 (a) 20 (b) 8 (c) 14 (d) −10 (e) −3 (f) 12 (g) −5 (h) 5

B7 (a) $2\frac{1}{2}$ (b) $2\frac{1}{4}$ (c) $3\frac{1}{5}$ (d) $3\frac{5}{6}$ (e) $\frac{5}{3}$ (f) $\frac{7}{2}$ (g) $\frac{12}{5}$ (h) $\frac{23}{4}$

B8 (a) $x + 5$ (b) $x - 8$ (c) $4x$ (d) $\frac{x}{6}$ (e) $10x$ (f) $2x - 3$ (g) $2(x + 9)$ (h) x^2

B9 (a) 13 (b) 4 (c) 20 (d) 12 (e) 11 (f) 16 (g) 28 (h) 3

B10 (a) $7a$ (b) $7k$ (c) $7t$ (d) $9xy$ (e) $3m + 2$ (f) $7g + 3h$ (g) $10n - 4$ (h) $3ab$

B11 (a) $8x$ (b) $30p$ (c) $6a$ (d) $12mn$ (e) ab (f) $14y^2$ (g) $15gh$ (h) $100d$

B12 (a) 42 (b) 13 (c) 40 (d) 77 (e) 2400 (f) 3000 (g) 216 (h) 78 (i) 124

B13 (a) 4 (b) 9 (c) 13 (d) 14 (e) 7 (f) 8 (g) 14 (h) 20

B14 (a) 4 (b) 5 (c) 3 (d) 6 (e) 10 (f) 17 (g) 14 (h) 10

Part C · Data and Probability

C1 Dots above each value – 0: 3 dots, 1: 8 dots, 2: 5 dots, 3: 2 dots, 4: 1 dot, 5: 1 dot (20 dots in total)

C2 (a) 20 (b) 3 + 8 = 11 (c) 2 + 1 + 1 = 4 (d) 5 + 2 + 1 + 1 = 9 (e) 8

C3 (a) mode 1, range 5 (b) mode 3, range 6 (c) modes 12 and 15, range 9 (d) no mode, range 8

C4 (a) yes, 25 (b) no outlier (c) yes, 2 (d) yes, 80

C5 (a) 5 (b) 5 (c) 20 ÷ 5 = 4 (d) 70 ÷ 5 = 14 (e) 12 ÷ 4 = 3 (f) 30 ÷ 6 = 5 (g) 10 ÷ 4 = 2.5

C6 (a) 5 (b) 6 (c) 4 (d) 5 (e) 6 (f) 3

C7 1. (a) 35 ÷ 7 = 5 (b) 5 (c) 48 ÷ 8 = 6 (d) halfway between 5 and 6 = 5.5. 2. (a) 15 ÷ 5 = 3 (b) 3 (c) 24 ÷ 6 = 4 (d) halfway between 3 and 4 = 3.5

C8 (a) 1 hour (b) 4 hours (c) Saturday (d) Tuesday (e) 23 hours (f) 4 hours

C9 (a) $\frac{6}{24} = \frac{1}{4}$ (b) $\frac{2}{24} = \frac{1}{12}$ (c) $\frac{4}{24} = \frac{1}{6}$ (d) $\frac{8}{24} = \frac{1}{3}$ (e) $\frac{45}{60} = \frac{3}{4}$ (f) $\frac{3}{7}$

C10 (a) {red, blue, green, orange} (b) {1, 2, 3, 4, 5} (c) {B, A, N} (d) {odd, even}

C11 (a) $\frac{4}{10} = \frac{2}{5}$ (b) $\frac{3}{10}$ (c) $\frac{2}{10} = \frac{1}{5}$ (d) $\frac{1}{10}$ (e) $\frac{7}{10}$ (f) 0 (g) $\frac{1}{6}$ (h) $\frac{3}{6} = \frac{1}{2}$

C12 (a) spinning red, green or orange; $\frac{7}{10}$ (b) spinning red, blue or green; $\frac{9}{10}$ (c) rolling 1, 2, 3, 4 or 5; $\frac{5}{6}$ (d) rolling an odd number; $\frac{1}{2}$ (e) tossing a tail; $\frac{1}{2}$ (f) $1 - 0.3 = 0.7$